

LAB 2

GPIO Interfacing- PART 1

Objectives:

WEEK 5

- How to create a C project in Keil uVision5
- To interface **LEDs and 7-Segment Displays** to the **GPIO** pins of **TI TIVA Launchpad** and write the required firmware

Material:

- **Keil uVision5 IDE**
- **TIVA LaunchPAD**
- **Common Anode 7-Segment Display** (available in your toolkit)
- **74LS247 IC** (BCD to 7-Segment Decoder/Driver IC, available in your toolkit)
- **Breadboard**
- **Jumper Wires**

Submission Instruction

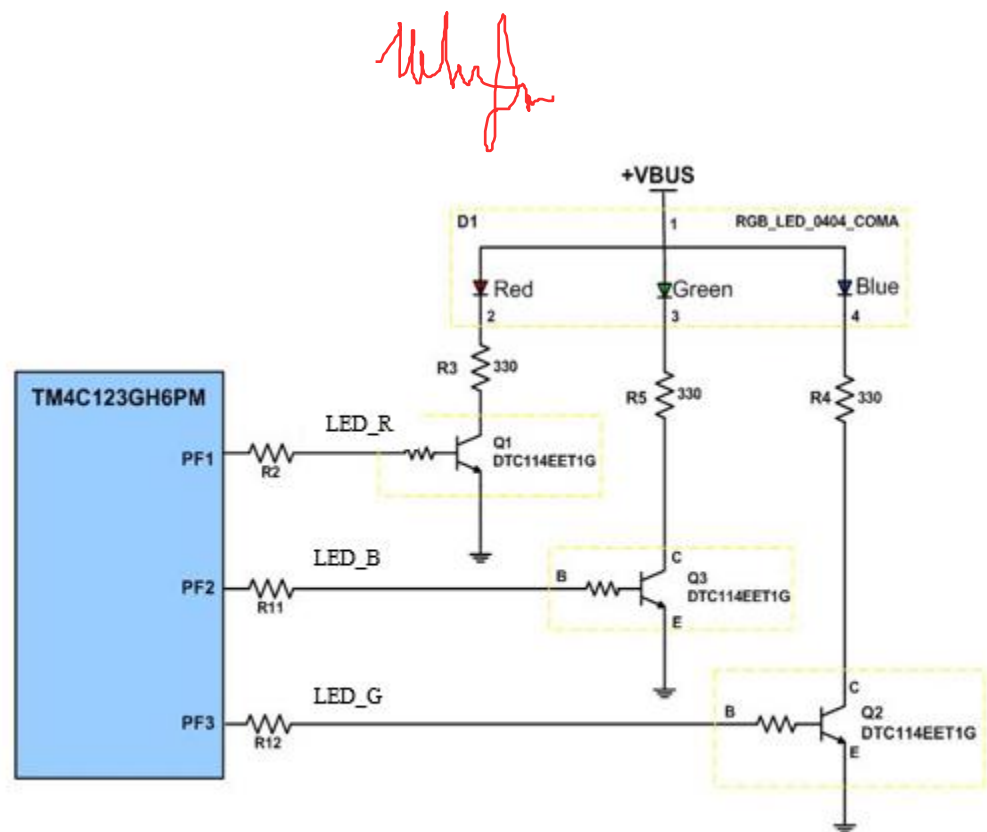
1. **Demonstrate** your work for each activity to your instructor. Missing the instructor signature results in **20% penalty**.
2. **Submit** a zipped file of your project folder onto the BB dropbox. Missing the zipped project file results in a **20% penalty**.
3. **Write** a lab report detailing the implementation and results. Include the picture of your circuit, screenshot of your code and results. Answer all the questions.

Lab Activity 1- Creating a C Project in Keil

- Follow the steps in [How to Create a C Project in Keil uVision](#), that is posted on BB under “Additional Resources” to create your first C program.
- Observe the **Red LED blinking every 0.5s** on the TIVA board.

Lab Activity 2- Control LEDs [20 Points]

Write a C program to turn on the Green LED on PF3 constantly and toggle red LED on PF1 for 1 second on and 2 second off infinitely.



LAB Activity 3- Interfacing with a 7-Segment Display [40 points]

Objective:

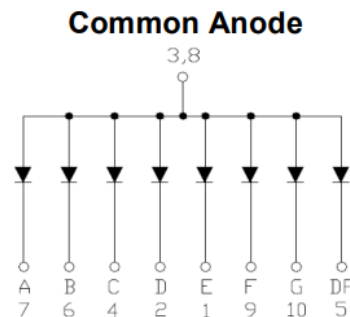
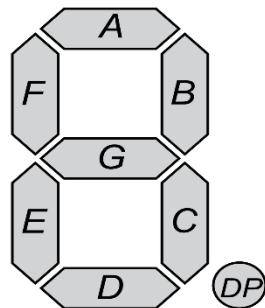
This lab aims to familiarize students with interfacing a Common Anode 7-segment display using the 74LS247 BCD to 7-segment decoder and the TI TIVA LaunchPad. Students will learn to configure GPIO Port B (PB0–PB3) to send BCD values and display numbers 0-9 sequentially. By the end of this lab, students will be able to write a C program to control the display and implement a 1-second counter using a delay loop or timer.

Background

A **Common Anode** 7-segment display has all **anodes connected to VCC (3.3V)**, and segments are **turned ON by driving them LOW (0)**.

Pin Mapping for a Common Anode 7-Segment Display:

Table 1 7-Segment Display Pin Diagram



Segment	Pin on Display
A	1
B	2
C	4
D	5
E	6
F	7
G	9
DP	3
VCC	8,10

Each segment **turns ON when driven LOW (0)**.

The **74LS247** is a BCD to 7-segment decoder/driver designed for **Common Anode** displays. It takes a 4-bit Binary Coded Decimal (BCD) input on A, B, C, D pins and decodes it to drive a 7-segment display, simplifying the process of displaying decimal digits.

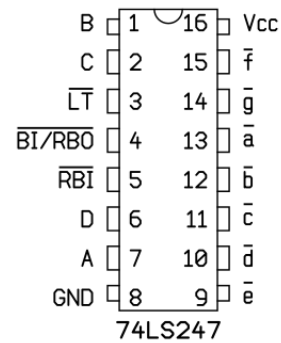


Table 2 represents the **truth table** for a **7-segment display** controlled by a **BCD (Binary-Coded Decimal) to 7-segment decoder** (e.g., 74LS247). It shows how the **4-bit binary inputs (D, C, B, A)** control the **7-segment outputs (a to g)** to display digits **0-9**.

Table 2 Function Table

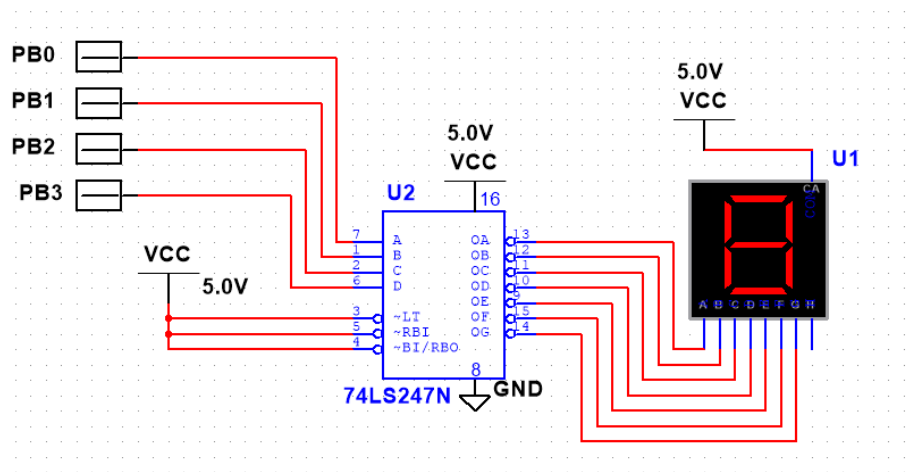
Inputs				Outputs							Decimal Value Displayed on 7-Seg
D	C	B	A	a	b	c	d	e	f	g	
0	0	0	0	ON	ON	ON	ON	ON	ON	OFF	0
0	0	0	1	OFF	ON	ON	OFF	OFF	OFF	OFF	1
0	0	1	0	ON	ON	OFF	ON	ON	OFF	ON	2
0	0	1	1	ON	ON	ON	ON	OFF	OFF	ON	3
0	1	0	0	OFF	ON	ON	OFF	OFF	ON	ON	4
0	1	0	1	ON	OFF	ON	ON	OFF	ON	ON	5
0	1	1	0	ON	OFF	ON	ON	ON	ON	ON	6
0	1	1	1	ON	ON	ON	OFF	OFF	OFF	OFF	7
1	0	0	0	ON	ON	ON	ON	ON	ON	ON	8
1	0	0	1	ON	ON	ON	ON	OFF	ON	ON	9

LAB Instructions

Step 1- Circuit Connection

The circuit connection for interfacing Common Anode 7-segment display with TIVA Launchpad via BCD to 7-segment decoder/driver IC is shown in the picture below. In this circuit, PB0 to PB3 are the first four pins on PORT B of TIVA LaunchPad.

Note: Ensure all ground connections are common between the TIVA LaunchPad, 74LS247, and the 7-segment display.



Step 2- Programming the TIVA Launchpad to display a number between 0 and 9 on 7-segment display

1. Initialize GPIO Port B:

- Enable Clock for Port B
- Configure PB0–PB3 as outputs for BCD data.
- Enable Digital Function on PB0-PB3

2. Send BCD Data:

- To display a digit, send the corresponding BCD value to PB0–PB3.

For example, to display the digit 5:

BCD for 5 is 0101

Set PB3 = 0, PB2 = 1, PB1 = 0, PB0 = 1

Step 3- Create a Counter

- Write a program to display **digits 0-9** in sequence with 1-second intervals.

Questions [40 points]

Understanding Concepts [15 points]

1. How does a **Common Anode** 7-segment display differ from a **Common Cathode** display?
2. What is the advantage of using a 74LS247 over directly interfacing a 7-segment display with a microcontroller?
3. How did you implement the **1-second delay** in your counter program?

Debugging and Troubleshooting [10 points]

4. What were the most common issues you faced when interfacing the TIVA LaunchPad with the 7-segment display?
5. How did/could you verify that the correct BCD values were sent to the 74LS247?

Applications and Extension [15 points]

6. How could this knowledge be applied to real-world embedded systems applications?
7. If you were to expand this project, what additional features or components would you include?